

Using REDCap for Data Research and Collection

Presented by: Muhimbili Orthopaedic Institute (MOI) REDCap (Research Electronic Data Capture) is a secure, web-based platform used for building and managing online surveys and databases in research, healthcare, and education.

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What is REDCap?

• Secure, web-based application • Developed by Vanderbilt University • Used worldwide for data research and collection • Supports both single and multi-center studies

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Why Use REDCap?

• Ensures data security and confidentiality • Structured data collection and validation • Real-time multi-user data entry • Easy export to analysis tools (SPSS, Stata, R) • Reduces paperwork and errors

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REDCap in Research Workflow

1. Project Design – Create forms and surveys 2. Data Collection – Enter or import participant data 3. Data Management – Validate and monitor data 4. Analysis – Export data to analysis tools 5. Reporting – Generate dashboards or reports

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Setting Up a REDCap Project

1. Log in to your REDCap system 2. Click '+ New Project' 3. Choose project type and enter details 4. Start from scratch, use a template, or import a data dictionary

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Designing Data Collection Instruments

• Use Online Designer for form creation • Field Types: text, choice, checkbox, file upload • Branching logic and field validation • Required fields and auto-numbering

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Using Surveys

- Convert forms into surveys • Distribute via email, link, or QR code • Schedule survey invitations • Enable anonymous responses • View real-time results

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Data Entry and Management

- Manual or bulk data entry via CSV • Role-based user permissions • Data quality checks and audit trails

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Data Export and Analysis

- Export to SPSS, Stata, R, Excel, SAS • De-identified data export • Generate reports and dashboards

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Collaboration and Multi-site Studies

- Multi-user and multi-site support • Role-based access • Data Access Groups (DAGs) • Centralized monitoring

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REDCap in Our Institution

- Patient intake and clinical studies • Laboratory data tracking • Follow-up surveys • Educational research and analytics

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Best Practices

- ✓ Pilot test forms before data collection ✓ Train users on permissions and data entry ✓ Regular backups ✓ Use consistent variable names ✓ Regular data quality reviews

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Challenges and Solutions

Challenge: Complex logic → Test in dev mode
Challenge: Access control → Set permissions carefully
Challenge: Validation errors → Use field checks
Challenge: Offline work → Use REDCap Mobile App

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REDCap Mobile App

- Collect data offline and sync later
- Secure data encryption
- Ideal for fieldwork and surveys

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Summary

REDCap enables: • Secure, reliable data collection • Collaboration across departments • Improved data quality for research

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Questions & Discussion

Thank you! Any questions or demo requests?

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